AoI-Aware Multi-User Downlink Transmission Scheme for LEO Satellite Systems

Yeling Dai, Min Lin, *Member, IEEE*, Xuewen Wu, *Member, IEEE*, Xiaoyu Liu, and Prabhat K. Upadhyay, *Senior Member, IEEE*

***Abstract*—Due to the increasing demands of users for information freshness and constrained power resources of low-earth orbit (LEO) satellite, this letter proposes a multi-user downlink transmission scheme to achieve a trade-off between age of information (AoI) and transmit power for LEO satellite systems. To this end, a constrained optimization problem is first formulated to minimize the weighted sum of long-term average AoI and transmit power, subject to the constraints of the maximal time-averaged AoI, the limited number of scheduled users, and the transmit power budget. Then, with the help of the drift-plus-penalty minimization method, the long-term optimization problem is transformed into a single-slot problem, so that the complex problem is mathematically tractable. Moreover, an angular-based joint beamforming and multi-user scheduling strategy is proposed to address the optimization problem. Finally, simulations verify that the proposed transmission scheme outperforms benchmarks in balancing AoI and transmit power.**

***Index Terms*—Age of information, beamforming, low-earth orbit satellite systems, power consumption, trade-off.**

I. Introduction

LOW-earth orbit (LEO) satellite communication plays a significant role by providing seamless coverage and reliable connectivity, especially in remote areas [1], [2]. However, with the increasing number of satellite users and limited spectrum resources, numerous resource allocation schemes have been proposed to improve resource utilization. For example, the authors of [3] and [4] jointly optimized power allocation and beamforming to minimize total transmit power, providing valuable insights for transmission scheme design.

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In recent years, the stringent latency requirements of users in LEO satellite systems have made timely information transmission a critical challenge [5]. The age of information (AoI), which quantifies the time elapsed from information generation to successful reception, has emerged as a key information freshness metric [6]. Reducing AoI can be achieved by increasing the transmission frequency [7] or improving reliability within limited retransmissions [8], both at the cost of higher power consumption. Conversely, minimizing transmit power often leads to sparse transmissions and outdated information. This contradiction is particularly serious in LEO satellite systems with constrained power budgets and growing real-time demands, making joint AoI and transmit power optimization essential. Specifically, a long-term AoI minimization power allocation problem was formulated in [9], but serving all users simultaneously works well only for a small user set. With more users, joint user scheduling and resource allocation are necessary. Furthermore, beamforming can enhance signal strength and reliability, further reducing AoI. In this context, an AoI-aware energy efficiency resource allocation scheme was proposed in [10] using alternating optimization to determine scheduled users and beamforming vectors, though the high computational complexity limits real-time applicability for large user numbers.

Motivated by these observations, we propose a low-complexity multi-user downlink transmission scheme in LEO satellite systems that jointly considers timeliness and power consumption. Unlike existing work [7], [8], [9], dynamic beams are utilized to satisfy the timeliness requirements of different scheduled users. We first formulate a constrained optimization problem that minimizes the weighted sum of long-term average AoI and transmit power, subject to maximum time-average AoI, limited scheduled users, and power budget constraints. The long-term problem is then transformed into a single-slot problem via the drift-plus-penalty (DPP) minimization method. Angular-based beamforming design utilizing the Lagrangian multiplier method and multi-user scheduling strategy based on the update index are proposed to address the problem, yielding the AoI-aware multi-user downlink transmission (AMDT) scheme. Simulations verify that the proposed scheme effectively balances AoI and transmit power.

II. System Model

This letter investigates multi-user downlink transmission in a LEO satellite system, where M users, randomly distributed within a circular area of radius γ_0, have stringent real-time requirements measured by AoI. The satellite (SAT) employs a uniform planar array (UPA) with $N = N_x \times N_y$ elements

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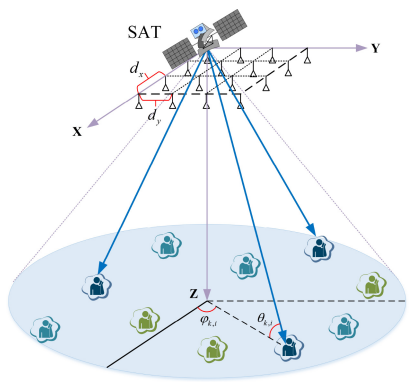

Fig. 1. An illustration of the considered system.

to implement space division multiple access (SDMA). Since the number of users exceeds the number of antennas and the limited degree of freedom, user grouping is necessary. Specifically, based on angular information, users with low channel correlation are grouped together to reduce intra-group interference. Thus, M users are divided into K groups, denoted by $\mathscr{S}_k (1 \leqslant k \leqslant K)$, with users in the same group marked by the same color in Fig. 1. Users within each group are served via SDMA, while users from different groups through time division multiple access.

A. Signal Model

At time slot t, the SAT employs beamforming technology to transmit signals to scheduled user group $\mathscr{S}_{k^*}$. Assuming the SAT transmits signal $x_{k^*,i}(t)$ satisfying $\mathbb{E}[|x_{k^*,i}(t)|^2] = 1$ to user $U_{k^*,i}$ with the transmit beamforming weight vector $\mathbf{w}_{k^*,i}(t) \in \mathbb{C}^{N\times 1}$, the transmit signal of SAT is expressed as

$$\mathbf{s}_{k^*}(t) = \sum_{i=1}^{|\mathscr{S}_{k^*}|} \mathbf{w}_{k^*,i}(t) x_{k^*,i}(t), \tag{1}$$

where $|\mathscr{S}_{k^*}|$ represents the number of users in group $\mathscr{S}_{k^*}$. The received signal at $U_{k^*,i}$ is then expressed as

$$y_{k^*,i}(t) = \mathbf{h}_{k^*,i}^H(t) \sum_{i=1}^{|\mathscr{S}_{k^*}|} \mathbf{w}_{k^*,i}(t) x_{k^*,i}(t) + n_{k^*,i}(t), \tag{2}$$

where $n_{k^*,i}(t)$ is the additive white Gaussian noise with mean zero and variance $\sigma^2_{k^*,i}$. $\mathbf{h}_{k^*,i}(t) \in \mathbb{C}^{N\times 1}$ is the channel vector between SAT and $U_{k^*,i}$, which can be modeled as [11]

$$\mathbf{h}_{k^*,i}(t) = \sqrt{L_{k^*,i}} \rho_{k^*,i}(t) \mathbf{a}\big(\theta_{k^*,i}, \varphi_{k^*,i}\big) \exp\{j2\pi t v_{k^*,i}\}, \tag{3}$$

where $L_{k^*,i} = (c/4\pi d_{k^*,i} f_c)^2$ denotes the path loss from SAT to $U_{k^*,i}$, in which c is the light speed, f_c is the carrier frequency, and $d_{k^*,i}$ is the distance between SAT and $U_{k^*,i}$. $|\rho_{k^*,i}(t)|$ is a random variable that obeys the Shadowed-Rician distribution with parameter $m_{k^*,i}$, $b_{k^*,i}$ and $\Omega_{k^*,i}$. Furthermore, due to the employment of a UPA at the SAT, the array steering vector $\mathbf{a}(\theta_{k^*,i}, \varphi_{k^*,i})$ can be formulated as [11]

$$\mathbf{a}\big(\theta_{k^*,i}, \varphi_{k^*,i}\big) = \mathbf{a}_x\big(\theta_{k^*,i}, \varphi_{k^*,i}\big) \otimes \mathbf{a}_y\big(\theta_{k^*,i}, \varphi_{k^*,i}\big), \tag{4}$$

where $\otimes$ denotes the Kronecker product. $\mathbf{a}_x(\theta_{k^*,i}, \varphi_{k^*,i})$ and $\mathbf{a}_y(\theta_{k^*,i}, \varphi_{k^*,i})$ are the steering vectors alongside X-axis and Y-axis, respectively, which are given by

$$\mathbf{a}_x\big(\theta_{k^*,i}, \varphi_{k^*,i}\big) = \Big[1, e^{j\kappa d_x \omega_{x,k^*,i}}, \ldots, e^{j(N_x-1)\kappa d_x \omega_{x,k^*,i}}\Big]^T, \tag{5}$$

$$\mathbf{a}_y\big(\theta_{k^*,i}, \varphi_{k^*,i}\big) = \Big[1, e^{j\kappa d_y \omega_{y,k^*,i}}, \ldots, e^{j(N_y-1)\kappa d_y \omega_{y,k^*,i}}\Big]^T, \tag{6}$$

where $\omega_{x,k^*,i} = \cos\theta_{k^*,i}\cos\varphi_{k^*,i}$ and $\omega_{y,k^*,i} = \cos\theta_{k^*,i}\sin\varphi_{k^*,i}$. $\kappa = 2\pi/\lambda$ is the wavenumber. d_x and d_y are the array inter-element spacing along X-axis and Y-axis. $\theta_{k^*,i} \in [0, \pi/2)$ and $\varphi_{k^*,i} \in [0, 2\pi)$ are the elevation and azimuth angle-of-arrival at Uk^*, i.

In (3), $v_{k^*,i}$ denotes the Doppler shift due to SAT mobility. Using self-positioning and ephemeris data, Doppler shift can be pre-compensated, effectively mitigating its impact [11]. Therefore, $\mathbf{h}_{k^*,i}(t)$ is simplified as

$$\mathbf{h}_{k^*,i}(t) = \sqrt{L_{k^*,i}} \rho_{k^*,i}(t) \mathbf{a}\big(\theta_{k^*,i}, \varphi_{k^*,i}\big). \tag{7}$$

The output signal-to-interference-plus-noise ratio (SINR) of $U_{k^*,i}$ is given by

$$\gamma_{k^*,i}(t) = \frac{\left|\mathbf{h}_{k^*,i}^H(t)\mathbf{w}_{k^*,i}\right|^2}{\sum_{j=1,j\neq i}^{|\mathscr{S}_{k^*}|} \left|\mathbf{h}_{k^*,i}^H(t)\mathbf{w}_{k^*,j}(t)\right|^2 + \sigma^2_{k^*,i}}. \tag{8}$$

According to (8), the achievable rate of $U_{k^*,i}$ is expressed as $R_{k^*,i}(t) = \log_2(1 + \gamma_{k^*,i}(t))$.

B. AoI Evolution Model

To capture real-time information delivery, we define AoI $A_{k,i}(t)$ for $U_{k,i}$. The scheduling indicator $\delta_k(t)$ denotes whether group $\mathscr{S}_k$ is scheduled at time slot t. Specifically, if $\mathscr{S}_k$ is scheduled, it is denoted by $\delta_k(t) = 1$, and otherwise $\delta_k(t) = 0$. Moreover, if the signal of scheduled user can be successfully transmitted, the AoI will reset to one at $t + 1$. Otherwise, it will increment by one [5]. Therefore, the evolution of $A_{k,i}(t)$ is characterized as

$$A_{k,i}(t+1) = \begin{cases} 1, & \delta_k(t)\zeta_{k,i}(t) = 1 \\ A_{k,i}(t) + 1, & \delta_k(t)\zeta_{k,i}(t) = 0, \end{cases} \tag{9}$$

where the successful transmission variable $\zeta_{k,i}(t)$ denotes the result of signal transmission. Specifically, if the achievable rate $R_{k,i}$ exceeds the rate threshold $R^{th}_{k,i}$, the signal of $U_{k,i}$ can be successfully transmitted at time slot t, indicated by $\zeta_{k,i}(t) = 1$, and otherwise $\zeta_{k,i}(t) = 0$.

III. Proposed Downlink Transmission Scheme

Minimizing AoI is crucial in LEO satellite systems due to growing real-time transmission demands, though it typically increases power consumption [7], [8]. With limited LEO satellite power budget it is essential to design a transmission scheme to balance AoI and transmit power. Furthermore, to capture long-term characteristics, we define the long-term average AoI $\bar{A}$ and transmit power $\bar{P}$ as

$$\bar{A} \triangleq \lim_{T\to\infty} \frac{1}{MT} \sum_{t=0}^{T-1} \sum_{k=1}^{K} \sum_{i=1}^{|\mathscr{S}_k|} A_{k,i}(t),$$

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$$\bar{P} \triangleq \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \sum_{k=1}^{K} \sum_{i=1}^{|\mathscr{S}_k|} \delta_k(t) \left\| \mathbf{w}_{k,i}(t) \right\|^2, \tag{10}$$

where T denotes the total time slots. We then formulate a constrained optimization problem to minimize the weighted sum of $\bar{A}$ and $\bar{P}$, by jointly optimizing $\delta_k(t)$ and $\mathbf{w}_{k,i}(t)$ at each time slot, which is mathematically formulated as

$$\min_{\delta_k(t),\mathbf{w}_{k,i}(t)} \bar{\Theta} = \alpha \bar{A} + \beta(1-\alpha)\bar{P} \tag{11a}$$

$$\text{s.t.} \quad \bar{A}_{k,i} \le \Delta_{k,i}^{\max}, \forall i, \forall k, \tag{11b}$$

$$\delta_k(t) \in \{0,1\}, \forall k, \tag{11c}$$

$$\sum_{k=1}^{K} \delta_k(t) = 1, \tag{11d}$$

$$\sum_{k=1}^{K} \sum_{i=1}^{|\mathscr{S}_k|} \delta_k(t) \left\| \mathbf{w}_{k,i}(t) \right\|^2 \le P^{\max}, \tag{11e}$$

where $\alpha \in [0,1]$ weights the relative importance of $\bar{A}$ and $\bar{P}$, and $\beta \ge 0$ normalizes their magnitudes. Furthermore, (11b) limits the time-averaged AoI of $U_{k,i}$ to $\Delta_{k,i}^{\max}$. Eq. (11c) represents the binary characteristic of $\delta_k(t)$. Eq. (11d) indicates that only one user group can be scheduled at time slot t. Eq. (11e) captures the transmit power budget $P^{\max}$ of SAT. (11) excludes explicit rate constraints to align with the AoI evolution model and maintain generality under stochastic transmission outcomes.

A. Problem Transformation Based on DPP Method

Directly solving problem (11) is difficult due to the long-term objective (11a) and constraint (11b). To simplify, we apply the DPP minimization method [12] and introduce the virtual queue $Y_{k,i}(t)$ for $U_{k,i}$, transforming the time-averaged AoI constraint (11b) into a queue stability constraint, which is updated as [10]

$$Y_{k,i}(t+1) = \max\Big\{ Y_{k,i}(t) + A_{k,i}(t+1) - \Delta_{k,i}^{\max}, 0 \Big\}. \tag{12}$$

Let $\mathbf{Y}(t)$ denote the vector of all virtual queues. Stability of $\mathbf{Y}(t)$ ensures constraint (12)b. The quadratic Lyapunov function $L(\mathbf{Y}(t))$ is then defined to measure queue congestion

$$L(\mathbf{Y}(t)) \triangleq \frac{1}{2} \sum_{k=1}^{K} \sum_{i=1}^{|\mathscr{S}_k|} Y_{k,i}^2(t). \tag{13}$$

Therefore, $\mathbf{Y}(t)$ can be stabilized by minimizing the change in $L(\mathbf{Y}(t))$ between consecutive time slots, which is defined as the Lyapunov drift $\Delta(\mathbf{Y}(t))$, and is expressed as

$$\Delta(\mathbf{Y}(t)) \triangleq L(\mathbf{Y}(t+1)) - L(\mathbf{Y}(t)). \tag{14}$$

Furthermore, the penalty term $\Theta(t)$ is defined as

$$\Theta(t) = \sum_{k=1}^{K} \sum_{i=1}^{|\mathscr{S}_k|} \Big[\alpha A_{k,i}(t) + \beta(1-\alpha)\delta_k(t) \left\| \mathbf{w}_{k,i}(t) \right\|^2 \Big]. \tag{15}$$

Under the DPP framework, problem (11) is equivalently transformed to minimize the upper bound of $\Delta(\mathbf{Y}(t)) + V\Theta(t)$ at each time slot, where $V \ge 0$ adjusts the objective weight. Using the inequality $(\max\{a+b-c,0\})^2 \le a^2 + b^2 + c^2 + 2a(b-c)$ for any $a \ge 0$, $b \ge 0$, and $c \ge 0$ [13], the upper bound of $\Delta(\mathbf{Y}(t)) + V\Theta(t)$ can be expressed as

$$\Delta(\mathbf{Y}(t)) + V\Theta(t) \le U + O^{\text{upper}}, \tag{16}$$

where $U = \sum_{k=1}^{K} \sum_{i=1}^{|\mathscr{S}_k|} \frac{1}{2}(A_{k,i}(t)+1)^2 + Y_{k,i}(t)(A_{k,i}(t) + 1 - \Delta_{k,i}^{\max}) + \frac{\alpha V}{M} A_{k,i}(t)$ is a constant at time slot t and O^{upper} is expressed as

$$O^{\text{upper}} = \sum_{k=1}^{K} \sum_{i=1}^{|\mathscr{S}_k|} \delta_k(t) \Big[\frac{1}{2} \zeta_{k,i}(t) \Big(1 - \big(A_{k,i}(t)+1\big)^2 - 2Y_{k,i}(t)A_{k,i}(t) \Big) + V\beta(1-\alpha) \left\| \mathbf{w}_{k,i}(t) \right\|^2 \Big]. \tag{17}$$

Problem (11) is transformed to minimize O^{upper} as

$$\min_{\delta_k(t),\mathbf{w}_{k,i}(t)} O^{\text{upper}} \quad \text{s.t (11c)–(11e).} \tag{18}$$

It is evident that due to the stability guarantee of the DPP minimization method, the optimal solution obtained by solving problem (18) at each time slot is equivalent to the global optimal solution of the original problem (11).

B. Joint Beamforming and Multi-User Scheduling Strategy

Problem (18) remains challenging to solve due to its mixed-integer programming property, involving binary variables $\delta_k(t)$ and continuous variables $\mathbf{w}_{k,i}(t)$. To reduce complexity, it is further decomposed into two independent subproblems: beamforming subproblem involving $\mathbf{w}_{k,i}(t)$

$$P1: \min_{\mathbf{w}_{k,i}(t)} O^{\text{upper}} \tag{19a}$$

$$\text{s.t.} \quad (11e), \tag{19b}$$

$$R_{k,i}(t) \ge R_{k,i}^{th}, \forall i \in \mathscr{S}_k, \tag{19c}$$

and scheduling subproblem involving $\delta_k(t)$

$$P2: \min_{\delta_k(t)} O^{\text{upper}} \tag{20a}$$

$$\text{s.t (11c)–(11d),} \tag{20b}$$

where (19c) ensures successful transmission for all users in $\mathscr{S}_k$, i.e., $\zeta_{k,i}(t) = 1, \forall i \in \mathscr{S}_k$, and further minimizes the O^{upper}. Only users with sufficiently good channel conditions are scheduled, avoiding failed transmissions and unexpected AoI increase. While slightly tightening the feasible region relative to (18), the optimal solution remains unchanged.

For the beamforming subproblem (19), we propose an angular-based beamforming scheme. Firstly, to reduce intra-group interference and achieve SDMA effectively, users with low channel correlation are clustered together. Unlike existing algorithms relying on instantaneous channel state information [14] and require frequent group updates, we exploit slowly-varying angular information, quasi-static over multiple slots [15], allowing stable grouping. Specifically,

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given angle θ_i and φ_i, the channel correlation c_{ij} ($1 \le i \le M$, $1 \le j \le M$) between U_i and U_j is expressed as

$$\begin{aligned} c_{ij} &= \mathbf{a}^H(\theta_i,\varphi_i)\mathbf{a}(\theta_j,\varphi_j)/\left(\left\|\mathbf{a}^H(\theta_i,\varphi_i)\right\| \cdot \left\|\mathbf{a}(\theta_j,\varphi_j)\right\|\right) \\ &= \frac{1}{N}\sum_{n_y=1}^{N_y}\sum_{n_x=1}^{N_x} \exp\big(j\kappa d_x(n_x-1)(\omega_{x,j}-\omega_{x,i})\big) \\ &\quad \times \exp\big(j\kappa d_y(n_y-1)(\omega_{y,j}-\omega_{y,i})\big), \end{aligned} \tag{21}$$

where smaller values of c_{ij} indicate lower channel correlation between U_i and U_j. To reduce intra-group interference, the channel correlations of users within each group must satisfy

$$c_{ij} < \varepsilon, \forall i,j \in \mathscr{S}_k, 1 \le k \le K, \tag{22}$$

where ε serves as a predefined orthogonality threshold.

The user grouping procedure is as follows. First, select the user U_{m^*} with the highest channel gain for each of the K initial groups. Then, select the next user U_{m^*} with the highest channel gain from the remaining users, calculate c_{m^*j} between U_{m^*} and all existing grouped users in $\mathscr{S}_k$, and identify the group $\mathscr{S}_{\hat{k}}$ with the lowest correlation, i.e., $\hat{k} = \arg\min_{1\le k\le K}(\max_{j\in\mathscr{S}_k} c_{m^*j})$. If $\max_{j\in\mathscr{S}_{\hat{k}}} c_{m^*j} < \varepsilon$, U_{m^*} will be assigned to $\mathscr{S}_{\hat{k}}$. Otherwise, a new group will be created specifically for U_{m^*}. Continue this process until all users are grouped. Ultimately, we can obtain the set of user groups $\mathscr{S}_k$.

For each group $\mathscr{S}_k$, the subproblem (27) is solved via the Lagrange multiplier method. Introducing multipliers $\lambda(t)$ and $\mu_i(t)$ to relax constraints (19b) and (19c), the Lagrangian function is obtained as follows

$$\begin{aligned} &\mathscr{L}\big(\mathbf{w}_{k,i}(t),\lambda(t),\mu_i(t)\big) = O^{\text{upper}} \\ &+\lambda(t)\left(\sum_{i=1}^{|\mathscr{S}_k|}\left\|\mathbf{w}_{k,i}(t)\right\|^2 - P^{\max}\right) + \sum_{i=1}^{|\mathscr{S}_k|}\mu_i(t)\Big(\gamma_{k,i}^{th} - \gamma_{k,i}(t)\Big), \end{aligned} \tag{23}$$

where $\gamma_{k,i}^{th} = 2^{R_{k,i}^{th}} - 1$. The optimal solution can be obtained by solving the following problem

$$\min_{\mathbf{w}_{k,i}(t)} \max_{\lambda(t),\mu_i(t)} \mathscr{L}\big(\mathbf{w}_{k,i}(t),\lambda(t),\mu_i(t)\big). \tag{24}$$

Specifically, leveraging Karush-Kuhn-Tucker conditions, optimal beamforming vectors can be calculated with optimal Lagrange multipliers $\lambda(t)$ and $\mu_i(t)$. By taking the partial derivative of the Lagrange function (24), $\mathbf{w}_{k,i}(t)$ is the unique solution to the $\partial(\mathscr{L}(\mathbf{w}_{k,i}(t),\lambda(t),\mu_i(t)))/\partial(\mathbf{w}_{k,i}(t)) = 0$. On the other hand, the optimal Lagrange multiples is iteratively updated via the subgradient projection method [16]

$$\begin{aligned} \lambda_{n+1}(t) &= \left[\lambda_n(t) - \eta_n(t)\left(P^{\max} - \sum_{i=1}^{|\mathscr{S}_k|}\left\|\mathbf{w}_{k,i}(t)\right\|^2\right)\right]^+ \\ \mu_{i,n+1}(t)) &= \left[\mu_{i,n}(t) - \eta_{i,n}(t)\Big(\gamma_{k,i}(t) - \gamma_{k,i}^{th}\Big)\right]^+, \forall i \in |\mathscr{S}_k|, \end{aligned} \tag{25}$$

where $[x]^+ = \max(0,x)$. $\eta_n(t)$ and $\eta_{i,n}(t)$ are the positive step sizes used in iterative updates of the Lagrange multipliers.

For the scheduling subproblem (20), since the objective function O^{upper} is a linear combination of scheduling indicator $\delta_k(t)$, the solution is relatively straightforward. Specifically, the update index $J_{k,i}(t)$ for $U_{k,i}$ is defined as

$$\begin{aligned} J_{k,i}(t) \triangleq \frac{1}{2}\zeta_{k,i}(t)\Big(1 - \big(A_{k,i}(t)+1\big)^2 - 2\,Y_{k,i}(t)A_{k,i}(t)\Big) \\ + V\beta(1-\alpha)\left\|\mathbf{w}_{k,i}(t)\right\|^2. \end{aligned} \tag{26}$$

This index takes into account both the AoI and transmit power. By exhausting all groups, the scheduled user group $\mathscr{S}_{k^*}$ at each time slot is determined by

$$k^* = \arg\min_{1\le k\le K} \sum_{i=1}^{|\mathscr{S}_k|} J_{k,i}(t). \tag{27}$$

The scheduling indicators are then updated as $\delta_{k^*}(t) = 1$ and $\delta_j(t) = 0, j \ne k^*$. The AoI and virtual queues are updated based on (9) and (12) to solve problem for next time slot, forming the proposed AMDT scheme in Algorithm 1, which has an outer loop over time slots and an inner loop over user groups. The inner-loop complexity scales with the group size and the number of floating point of operations (FLOP) is around 36 [13]. Therefore, the total computational complexity is $O(T\sum_{k=1}^{K} 36|\mathscr{S}_k|\mu^2)$ FLOPs.

Algorithm 1: Proposed AMDT Scheme

Input : $\{\lambda_n(t), \mu_{i,n}(t), \eta_n(t), \eta_{i,n}(t), T, \mu\}$;
Initialization: AoI $A_{k,i}(0)$ and virtual queue $Y_{k,i}(0)$;
Obtain the set of user groups $\mathscr{S}_k$;
for $t = 0$ *to* T **do**
  for $k = 1$ **to** K **do**
    while $|\lambda_{n+1}(t) - \lambda_n(t)| < \mu, |\mu_{i,n+1}(t) - \mu_{i,n}(t)| < \mu$ **do**
      Solve $\dfrac{\partial\Big(\mathscr{L}\Big(\mathbf{w}_{k,i}(t),\lambda(t),\mu_i(t)\Big)\Big)}{\partial\Big(\mathbf{w}_{k,i}(t)\Big)} = 0$;
      Update multipliers $\lambda_{n+1}(t)$ and $\mu_{i,n+1}(t)$;
      $n = n + 1$;
    end
    Calculate the update index $J_{k,i}(t), \forall i \in \mathscr{S}_k$;
  end
  Determine k^* through $k^* = \arg\min_{1\leqslant k\leqslant K} \sum_{i=1}^{|\mathscr{S}_k|} J_{k,i}(t)$;
  Update $\delta_{k^*}(t) = 1$ and $\delta_j(t) = 0, j \ne k^*$;
  Update AoI $A_{k,i}(t+1)$ and virtual queue $Y_{k,i}(t+1)$;
end
Output: The scheduling indicator $\delta_k(t)$ and the beamforming weight vectors $\mathbf{w}_{k,i}(t)$.

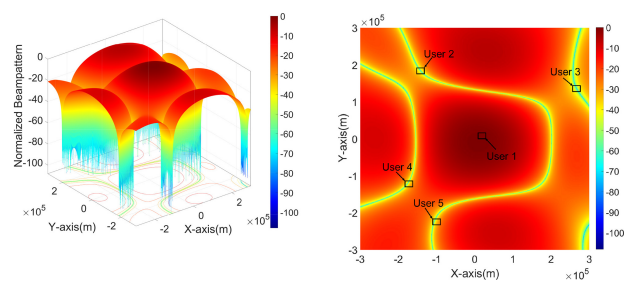

Fig. 2. Beampattern of $\mathbf{w}_1$ from 3D vision and vertical view.

IV. Simulation Results

We consider a LEO satellite system serving $M = 48$ users within the $\gamma_0 = 300$ km coverage radius, with satellite

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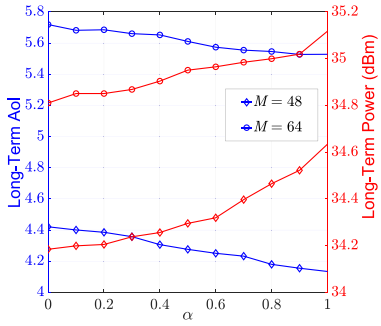

Fig. 3. Long-term average AoI and transmit power versus weight α.

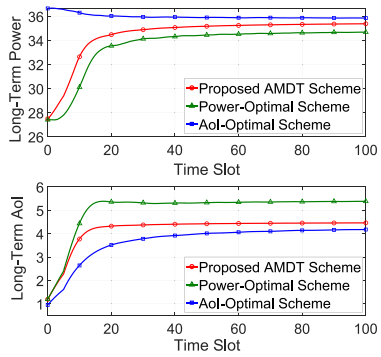

Fig. 4. Long-term average AoI and transmit power versus time slot.

altitude, carrier frequency, and antenna array size are set as $H = 500$ km, $f_c = 1.67$ GHz, and $N = 4 \times 4$. The channel experiences light shadowing, i.e., $\{m_{k,i}, b_{k,i}, \Omega_{k,i}\} = \{10, 0.126, 1.29\}$ [1]. Users are clustered into $K = 10$ groups via the proposed angle-based grouping method. The maximum time-averaged AoI, QoS threshold, maximum transmit power, and orthogonality threshold are set as $\Delta^{\max} = 5$, $R^{th} = 3$ bit/s/Hz, $P^{\max} = 40$ dBm and $\varepsilon = 0.4$ [3]. The weights are $\alpha = 0.5$, $\beta = 1$, and $V = 5$ is chosen empirically for AoIâĂŞpower trade-off. The step sizes, Lagrange multiples, and convergence threshold are initialized as $\eta_n(t) = \eta_{i,n}(t) = 0.01$, $\lambda_n(t) = \mu_{i,n}(t) = 1$ and $\mu = 10^{-3}$. The total time slot is $T = 100$. For performance benchmarking, we introduce two reference schemes: **Power-optimal scheme** minimizing the long-time transmit power (i.e., $\alpha = 0$), and **AoI-optimal scheme [9]** AoI (i.e., $\alpha = 1$), both under identical constraints to our proposed scheme.

Figure 2 illustrates the proposed scheme from 3D vision and vertical view of the normalized beampattern for the scheduled user U_1 at a time slot. The maximum beam gain is oriented towards U_1 and several nulls are formed in the regions of the other scheduled users, demonstrating that the proposed beamforming scheme is capable of meeting the QoS requirements of U_1 and limiting interference from other users.

Figure 3 shows the impact of weight α on long-term average AoI and transmit power under different M. The parameter α effectively balances information freshness and power consumption. Furthermore, system performance degrades as M increases due to longer scheduling cycles and higher power requirements for larger groups.

Figure 4 compares different schemes in terms of long-term average AoI and transmit power versus time slots. The proposed AMDT scheme achieves a superior performance trade-off between information freshness and power consumption. Specifically, the AoI-optimal scheme lowers AoI at the expense of higher power, while the power-optimal scheme reduces power but increases AoI.

V. Conclusion

This letter has proposed a multi-user downlink transmission scheme to guarantee information timeliness and reduce power consumption simultaneously in LEO satellite systems. Specifically, a constrained optimization problem minimizing the weighted sum of long-term average AoI and transmit power is solved via the proposed AMDT scheme, the effectiveness of which was validated through simulations.

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